

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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Case Study	MODULE LOAN: LOAN APPLICATION AND LOAN APPROVAL
System Name	KADA WEB-BASED COOPERATIVE SYSTEM
Group Name	ALPHATECH

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

DDL

Table Creation for Notifications

```
CREATE TABLE tb_notification (  
  n_id INT PRIMARY KEY AUTO_INCREMENT,  
  n_message TEXT NOT NULL,  
  n_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  n_loan_application_id INT NOT NULL  
);
```

```
mysql> CREATE TABLE tb_notification (  
  -> n_id INT PRIMARY KEY AUTO_INCREMENT,  
  -> n_message TEXT NOT NULL,  
  -> n_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  -> n_loan_application_id INT NOT NULL  
  -> );  
Query OK, 0 rows affected (0.01 sec)
```

Adding Foreign Key to Notification Table

```
ALTER TABLE tb_notification  
ADD CONSTRAINT fk_loan_application  
FOREIGN KEY (n_loan_application_id) REFERENCES  
tb_loan_application(la_id) ON DELETE CASCADE;
```

```
mysql> ALTER TABLE tb_notification  
  -> ADD CONSTRAINT fk_loan_application  
  -> FOREIGN KEY (n_loan_application_id) REFERENCES tb_loan_application(la_id) ON DELETE CASCADE;  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

DML

Insert Notification for Loan Application

```
INSERT INTO tb_notification (n_message, n_loan_application_id)  
VALUES ('Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3  
- 5 hari bekerja', 96);
```

```
mysql> INSERT INTO tb_notification (n_message, n_loan_application_id)
-> VALUES ('Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari
bekerja', 96);
Query OK, 1 row affected (0.00 sec)
```

Retrieve Latest Notification for Specific Loan Application

```
SELECT n_message
FROM tb_notification
WHERE n_loan_application_id = 96
ORDER BY n_id DESC LIMIT 1;
```

```
mysql> SELECT n_message
-> FROM tb_notification
-> WHERE n_loan_application_id = 96
-> ORDER BY n_id DESC LIMIT 1;
+-----+-----+
| n_message |
+-----+-----+
| Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja |
+-----+-----+
1 row in set (0.00 sec)
```

Count Notifications Grouped by Application Status

```
SELECT tb_appstatus.app_desc,
COUNT(tb_notification.n_id) AS total_notifications
FROM tb_appstatus
LEFT JOIN tb_loan_application ON tb_appstatus.app_id =
tb_loan_application.la_status
LEFT JOIN tb_notification ON tb_notification.n_loan_application_id =
tb_loan_application.la_id
GROUP BY tb_appstatus.app_desc;
```

```
mysql> SELECT tb_appstatus.app_desc,
-> COUNT(tb_notification.n_id) AS total_notifications
-> FROM tb_appstatus
-> LEFT JOIN tb_loan_application ON tb_appstatus.app_id = tb_loan_application.la_status
-> LEFT JOIN tb_notification ON tb_notification.n_loan_application_id = tb_loan_application.la_id
-> GROUP BY tb_appstatus.app_desc;
+-----+-----+
| app_desc | total_notifications |
+-----+-----+
| Dihantar | 2 |
| Diterima | 0 |
| Ditolak | 0 |
| Sedang Berjalan | 2 |
| Selesai | 0 |
+-----+-----+
5 rows in set (0.00 sec)
```

Count Notifications Grouped by Member ID

```
SELECT tb_loan_application.la_member_id, COUNT(tb_notification.n_id)
AS total_notifications
FROM tb_notification
JOIN tb_loan_application ON tb_notification.n_loan_application_id =
tb_loan_application.la_id
GROUP BY tb_loan_application.la_member_id;
```

```
mysql> SELECT tb_loan_application.la_member_id, COUNT(tb_notification.n_id) AS total_notifications
-> FROM tb_notification
-> JOIN tb_loan_application ON tb_notification.n_loan_application_id = tb_loan_application.la_id
-> GROUP BY tb_loan_application.la_member_id;
+-----+-----+
| la_member_id | total_notifications |
+-----+-----+
| 100          | 2                  |
| 150          | 2                  |
+-----+-----+
2 rows in set (0.00 sec)
```

Retrieve Recent Notifications for Specific Member

```
SELECT tb_notification.n_message, tb_notification.n_date
FROM tb_notification
LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id =
tb_loan_application.la_id
WHERE tb_loan_application.la_member_id = 100
ORDER BY tb_notification.n_date DESC
LIMIT 5;
```

```
mysql> SELECT tb_notification.n_message, tb_notification.n_date
-> FROM tb_notification
-> LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id = tb_loan_application.la_id
-> WHERE tb_loan_application.la_member_id = 100
-> ORDER BY tb_notification.n_date DESC
-> LIMIT 5;
+-----+-----+
| n_message | n_date |
+-----+-----+
| Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja | 2025-01-26 14:17:26 |
| Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja | 2025-01-26 13:37:57 |
+-----+-----+
2 rows in set (0.00 sec)
```

Retrieve Notifications for Rejected Applications

```
SELECT tb_loan_application.la_member_id, tb_notification.n_message,  
tb_notification.n_date  
FROM tb_notification  
LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id =  
tb_loan_application.la_id  
LEFT JOIN tb_appstatus ON tb_loan_application.la_status =  
tb_appstatus.app_id  
WHERE tb_appstatus.app_desc = 'Ditolak'  
ORDER BY tb_notification.n_date ASC;
```

```
mysql> SELECT tb_loan_application.la_member_id, tb_notification.n_message, tb_notification.n_date  
-> FROM tb_notification  
-> LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id = tb_loan_application.la_id  
-> LEFT JOIN tb_appstatus ON tb_loan_application.la_status = tb_appstatus.app_id  
-> WHERE tb_appstatus.app_desc = 'Ditolak'  
-> ORDER BY tb_notification.n_date ASC;  
+-----+-----+-----+  
| la_member_id | n_message | n_date |  
+-----+-----+-----+  
| 100 | Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja | 2025-01-26 14:17:26 |  
| 100 | Permohonan pembiayaan anda telah ditolak. Sila hubungi bahagian HR untuk maklumat lanjut. | 2025-01-26 14:55:52 |  
+-----+-----+-----+  
2 rows in set (0.00 sec)
```

Retrieve Notifications for Ongoing Applications

```
SELECT tb_loan_application.la_member_id, tb_notification.n_message,  
tb_notification.n_date  
FROM tb_notification  
LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id =  
tb_loan_application.la_id  
LEFT JOIN tb_appstatus ON tb_loan_application.la_status =  
tb_appstatus.app_id  
WHERE tb_appstatus.app_desc = 'Sedang Berjalan'  
ORDER BY tb_notification.n_date ASC;
```

```
mysql> SELECT tb_loan_application.la_member_id, tb_notification.n_message, tb_notification.n_date  
-> FROM tb_notification  
-> LEFT JOIN tb_loan_application ON tb_notification.n_loan_application_id = tb_loan_application.la_id  
-> LEFT JOIN tb_appstatus ON tb_loan_application.la_status = tb_appstatus.app_id  
-> WHERE tb_appstatus.app_desc = 'Sedang Berjalan'  
-> ORDER BY tb_notification.n_date ASC;  
+-----+-----+-----+  
| la_member_id | n_message | n_date |  
+-----+-----+-----+  
| 150 | Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja | 2025-01-26 13:23:51 |  
| 150 | Permohonan pembiayaan anda telah diluluskan. Sila semak akaun anda untuk jumlah pinjaman. | 2025-01-26 13:28:51 |  
+-----+-----+-----+  
2 rows in set (0.00 sec)
```

List Notifications with Loan Application Status

```
SELECT tb_notification.n_id AS notification_id, tb_notification.n_message
AS message, tb_appstatus.app_desc AS loan_application_status
FROM tb_notification
JOIN tb_loan_application ON tb_notification.n_loan_application_id =
tb_loan_application.la_id
JOIN tb_appstatus ON tb_loan_application.la_status = tb_appstatus.app_id;
```

```
mysql> SELECT tb_notification.n_id AS notification_id, tb_notification.n_message AS message, tb_appstatus.app_desc AS loan_application_status
-> FROM tb_notification
-> JOIN tb_loan_application ON tb_notification.n_loan_application_id = tb_loan_application.la_id
-> JOIN tb_appstatus ON tb_loan_application.la_status = tb_appstatus.app_id;
```

notification_id	message	loan_application_status
1	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Sedang Berjalan
2	Permohonan pembiayaan anda telah diluluskan. Sila semak akaun anda untuk jumlah pinjaman.	Sedang Berjalan
4	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Dihantar
23	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Ditolak
24	Permohonan pembiayaan anda telah ditolak. Sila hubungi bahagian HR untuk maklumat lanjut.	Ditolak

5 rows in set (0.00 sec)

3.2 Interface Implementation

Notification in member dashboard.

Koperasi Kakitangan KADA Kelantan Berhad

Notifications

- Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja 2025-01-26 14:17
- Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja 2025-01-26 13:37

Member Share Details

- Share Fee: **RM 300.00**
- Modal Fee: **RM 0.00**
- Fixed Savings: **RM 0.00**
- Charity Fund: **RM 0.00**
- Member Savings: **RM 0.00**

Loan Details

- Al-Rai
- Al-Inah
- Skim Khac
- Karnival Mucim

Notification details for admin to check.



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Muka Hadapan
- Senarai Akaun
- Senarai Permohonan Kualifikasi
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Perincian Laporan
- Log Out

Notification Dashboard

Notifications by Application Status

Status	Total Notifications
Dihantar	1
Diterima	0
Ditolak	2
Sedang Berjalan	2
Selesai	0

Notifications by Member

Member ID	Total Notifications
100	3
150	2

All Notifications with Loan Application Status

Notification ID	Message	Loan Application Status
1	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Sedang Berjalan
2	Permohonan pembiayaan anda telah diluluskan. Sila semak akaun anda untuk jumlah pinjaman.	Sedang Berjalan
4	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Dihantar
23	Permohonan pembiayaan anda telah dihantar. Sila tunggu sekitar 3 - 5 hari bekerja	Ditolak
24	Permohonan pembiayaan anda telah ditolak. Sila hubungi bahagian HR untuk maklumat lanjut.	Ditolak

Loan application form that insert the notification to the notification table after submit it.



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Dashboard
- Apply Loans
- Cetak Statement
- Log Out

Butir-Butir Penjamin

Penjamin 1

No. KP

Nama

No. PF

No. Anggota

Borang Tandatangan Penjamin (PDF)

Maksimum Saiz Fail: 1 MB.

Penjamin 2

No. KP

Nama

No. PF

No. Anggota

Borang Tandatangan Penjamin (PDF)

Maksimum Saiz Fail: 1 MB.

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

The Notification Module has increased the efficiency of the loan application process in the KADA Web-Based Cooperative System. This module solves the weaknesses identified in phase 1, such as member confusion regarding loan statuses and staff workload caused by too many inquiries. With features like email notifications and a tracking dashboard, this will improve user satisfaction and ensure better communication within the system. Overall, the Notification Module makes the process more clearer, easier to handle, and more user-friendly to use.

4.2 Reflection

Description:

The project was about improving the loan module by adding a Notification Module. This included sending automated emails and showing real-time updates on the dashboard. Our team worked together on different modules, and I was responsible for the loan module and its notification feature. My tasks included designing the system, creating the database, and suggesting SQL queries and interface ideas.

Feelings:

At first, I felt stressed because combining different modules into one system was hard. Working with the team was good, but managing time and coordinating tasks was difficult. In the end, I felt happy because we built these modules that solved a KADA problem.

Evaluation:

The project gave me a chance to use what I learned in class to solve SQL and other code problems. The biggest challenges were making sure all the modules worked well together and fixing issues during integration. Good communication, teamwork and consistent improvement of the system helped us succeed.

Analysis:

I chose the Loan Module to reflect on because it made the system more user-friendly and fixed an important problem like users don't know when the loan application is approved. This project taught me to focus on what users need and to make small improvements that help the overall design. I also learned how to handle complex technical tasks while thinking about the user experience.

Conclusion:

I now understand how to design database systems for real use cases. I learned that new features like notifications can improve user satisfaction and make the system more efficient. However, I still need to figure out how to improve the system's performance before we present the complete version to our client.

Self-Criticism:

For future projects, I want to manage my time better and set clearer goals. I also want to learn more about designing for users and combining different system parts smoothly. Getting help from experts during tough times also could make my work even better.